

# **Sri Lanka Telecom Customer Churn**

## **Prediction Web Application**

### **1. Problem Statement**

Customer churn is a major challenge in the telecommunications industry. In highly competitive markets like Sri Lanka, customers frequently switch between service providers due to pricing, network quality, or service experience. Acquiring new customers is significantly more expensive than retaining existing ones. Therefore, telecom companies need a predictive system to identify customers who are at high risk of leaving. This project aims to develop a machine learning-powered web application that predicts whether a telecom customer is likely to churn, enabling proactive retention strategies.

### **2. Dataset**

Data set Source: - (From keggel)  
*IBM Telco Customer Churn Dataset*

Data set link - <https://www.kaggle.com/datasets/yeanzc/telco-customer-churn-ibm-dataset>

This is US available Data set. This dataset is used to simulate a Sri Lankan telecom environment.

### **3. ML Model**

This is a Supervised Machine Learning.

Planned models:

- Logistic Regression
- Random Forest Classifier
- Decision Tree Classifier

### **4. Proposed Tech Stack**

ML Pipeline: - *Python, Pandas, NumPy, Scikit-learn, Matplotlib*

Backend: - *Flask*

Frontend: - *HTML, CSS, JavaScript*

Model Deployment: - *Joblib*